



RELEASE HISTORY

# PowerGlove Vision Changelog

Versioned features, fixes, security changes, and documentation updates.

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This file records user-visible PowerGlove Vision changes. The project follows [Keep a Changelog](#) categories and uses [Semantic Versioning](#). Git remains the authoritative record for line-level and file-level history.

## Unreleased

### Documentation

- Standardized source headers with each file's purpose, author, copyright, SPDX license identifier, local history, and links to the complete history.
- Documented public interfaces and non-obvious security, lifecycle, tracking, packaging, and rendering functions.
- Added this centralized project changelog and its print-ready PDF edition.
- Added an automated audit for required source headers, module descriptions, and production Python interface docstrings.
- Added a complete configuration reference covering JSON templates and active copies, gesture thresholds, manifests, RetroArch mapping, systemd units, generated state, secret handling, and the public ZIP location.
- Added GitHub Actions checks for supported Python versions, tests, source and documentation audits, syntax, PDF builds, and public-package verification.
- Added a security policy for private reporting, pairing and network boundaries, shared-token handling, shutdown permissions, and dependency integrity.
- Added a contributing guide for code style, tests, documentation, changelog, generated artifacts, commits, and pull requests.
- Added reusable documentation and App Lab package verification tools.

## 0.1.0 - 2026-09-03

### Added

- Camera-only Power Glove tracking on Arduino UNO Q with MediaPipe.
- Gesture profiles for Bad Street Brawler, Super Glove Ball, and cartridge-free Programs A-I.
- Authenticated UDP profile selection and virtual Linux gamepad output for RetroPie, including per-game runcommand hooks.
- Dashboard, live diagnostics, offline gesture lessons, configuration controls, camera recovery, controller start/stop controls, and UNO Q matrix feedback.
- Wi-Fi deployment, App Lab packaging, runtime-asset retrieval, branded PDF generation, and a fixed-purpose host shutdown helper.
- Project overview, field guide, quick reference, profile handbook, third-party component notice, screenshots, and reproducible build instructions.

### Changed

- Delayed RetroPie receiver startup until EmulationStation finishes its initial device scan, preventing the virtual controller from disrupting cabinet input.

- Made the controller start disarmed and kept tracking/dashboard operation alive when RetroPie or the camera temporarily becomes unavailable.
- Moved private device settings and downloaded model data outside the public App Lab package.
- Changed the public package to download Google's Hand Landmarker model on first launch and verify its pinned SHA-256 digest before installation.

## Fixed

- Recovered cleanly from USB camera disconnects and slow UVC camera wake-up.
- Kept the vision loop responsive through receiver, DNS, Wi-Fi, and mDNS loss.
- Corrected password pairing so credentials never appear in command arguments and the shared token is transferred and installed reliably.
- Corrected UNO Q dependency isolation, secure token upload, PDF builder file mode, landscape diagnostics layout, and cabinet launch integration.

## Security

- Added short-lived TLS pairing with certificate comparison, a physical single-use PIN, bounded handshakes, and restricted token-file permissions.
- Required confirmation and a fixed host-side request path for system shutdown.
- Added a third-party component notice covering licenses, provenance, pinned versions, checksums, and update procedure.